ZIYUE (ALVIN) LIU

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EDUCATION

University of California, Santa Barbara, *Santa Barbara, CA* — *Sep. 2023 – Apr. 2027*
Doctor of Philosophy in Computer Science, GPA: 4.00/4.00

University of California, Santa Barbara, *Santa Barbara, CA* — *Sep. 2021 – Mar. 2023*
Master of Arts in Statistics, GPA: 3.97/4.00

Nankai University — School of Mathematical Sciences, *Tianjin, China* — *Aug. 2016 – Jun. 2020*
Bachelor of Science in Statistics, GPA: 3.72/4.00

RESEARCH

Efficient and Resilient Pre-Training of Large Language Models — *Sep. 2023 – Now*
- Efficient model architecture, optimizer, and infrastructure on leadership-class HPC.
- Resilient pre-training system via fault-tolerant communication primitives for O(100k+) GPUs.

AI for EDA and Data Center Thermal Management — *Feb. 2022 – Now*
- Physics-driven operator learning for fast and trustworthy 3D-IC thermal simulation and optimization.
- AI surrogates of data center thermal/CFD simulations, integrated with Cadence Reality Digital Twin Platform.

SELECTED PUBLICATIONS

[1] **Z. Liu**, Z. Wang, R. Zhang, A. Maurya, H. Zhou, P. Hovland, S. Di, F. Cappello, B. Nicolae, and Z. Zhang. **"ReCoVer: Resilient LLM Pre-Training System via Fault-Tolerant Collective and Versatile Workload ."** NeurIPS, 2026.

[2] **Z. Liu**, R. Zhang, Z. Wang, Y. Zhao, Y. Su, Z. Yang and Z. Zhang. **"Muon^2: Boosting Muon via Adaptive Second-Moment Preconditioning."** EMNLP **oral**, 2026.

[3] **Z. Liu***, R. Zhang*, Z. Wang*, M. Yan, Z. Yang, P. Hovland, B. Nicolae, F. Cappello, S. Tang and Z. Zhang. **"CoLA: Compute-Efficient Pre-Training of LLMs via Low-Rank Activation."** EMNLP **Oral**, 2025. (*Equal contribution)

[4] Z. Wang*, **Z. Liu***, R. Zhang, A. Maurya, B. Nicolae, P. Hovland, F. Cappello, and Z. Zhang. **"BOOST: BOttleneck-Optimized Scalable Training Framework for Low-Rank Large Language Models."** MLSys, 2026. (*Equal contribution)

[5] R. Zhang*, **Z. Liu***, Z. Wang and Z. Zhang. **"LaX: Boosting Low-Rank Training of Foundation Models via Latent Crossing."** NeurIPS, 2025. (*Equal contribution)

[6] **Z. Liu**, Y. Li, J. Hu, X. Yu, S. Shiau, Xin Ai, Z. Zeng, and Z. Zhang. **"DeepOHeat: Operator Learning-based Ultra-fast Thermal Simulation in 3D-IC Design."** Design Automation Conference (DAC), 2023.

[7] X. Yu, **Z. Liu**, H. Li, Y. Li, Xin Ai, Z. Zeng, I. Young and Z. Zhang. **"DeepOHeat-v1: Efficient Operator Learning for Fast and Trustworthy Thermal Simulation and Optimization in 3D-IC Design."** IEEE Trans. Components, Packaging and Manufacturing Technology, 2026.

[8] Z. Yang, **Z. Liu**, S. Choudhary, X. Xie, C. Gao, S. Kunzmann and Z. Zhang. **"CoMERA: Computing- and Memory-Efficient Training via Rank-Adaptive Tensor Optimization."** NeurIPS, 2024.

INTERNSHIPS

Google Cloud AI, *Sunnyvale, CA*
Software Engineer Intern: Efficient MoE adapter tuning and multi-tenant serving — *June. 2026 – Sep. 2026*

Argonne National Laboratory, *Lemont, IL*
Visiting Student: Fault-tolerant LLM pre-training — *April. 2025 – Sep. 2025*

Cadence Design Systems, *Austin, TX*
Research Intern: Data-driven modeling of real-world data center thermal/CFD simulations — *Jun. 2024 – Sep. 2024*
Research Intern: Data-driven modeling of large-scale automotive aerodynamic simulations — *Apr. 2023 – Sep. 2023*
Research Intern: Physics-informed operator learning for 3D-IC thermal simulations — *Jun. 2022 – Sep. 2022*